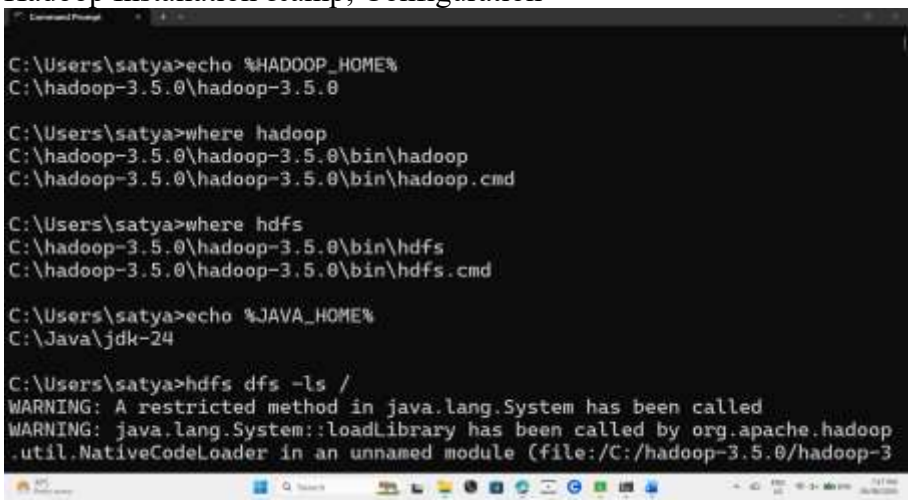
Practical 2

Aim: Hadoop Installation, Configuration and HDFS Operations on Windows/Linux.

Problem definition: You have recently joined DataAnalytics Inc. as a Hadoop Administrator. The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS).
As a Hadoop Administrator, your responsibility is to install and configure Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations.
Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently.
Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

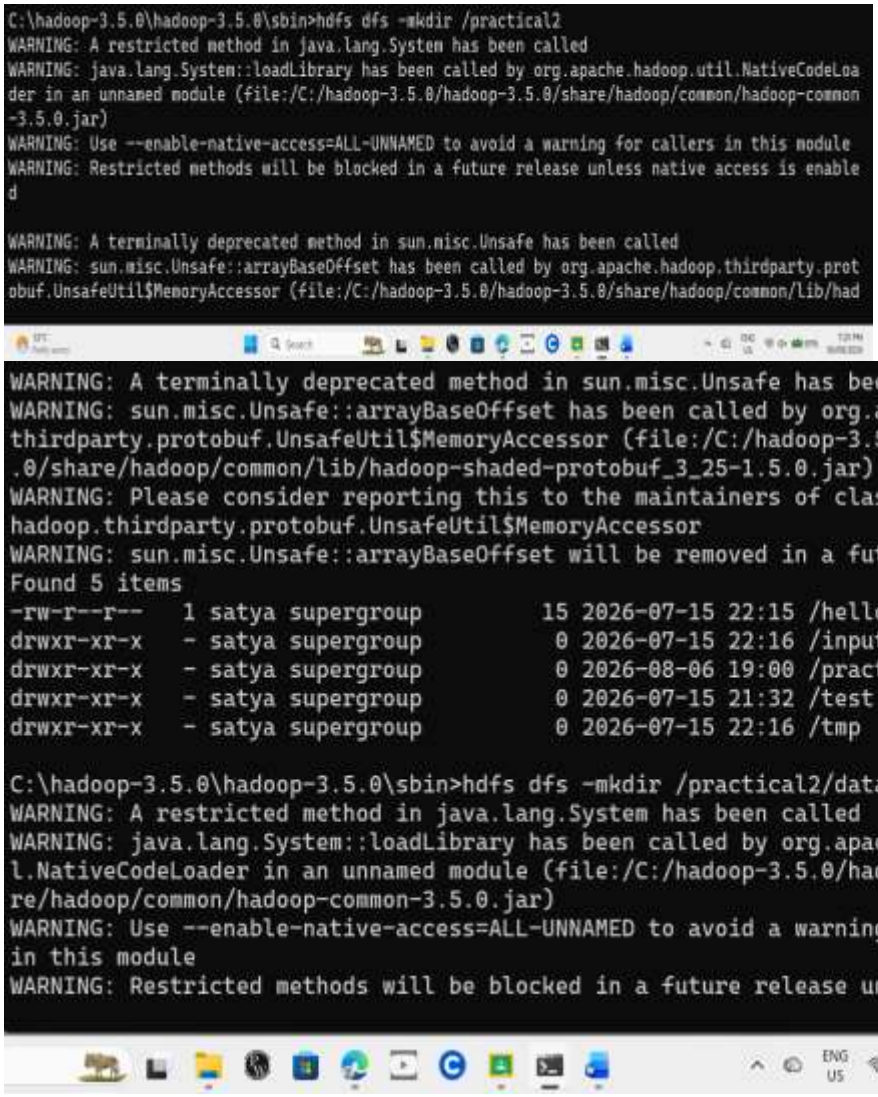
Observation / Output:
Tasks:
Hadoop Installation & Configuration



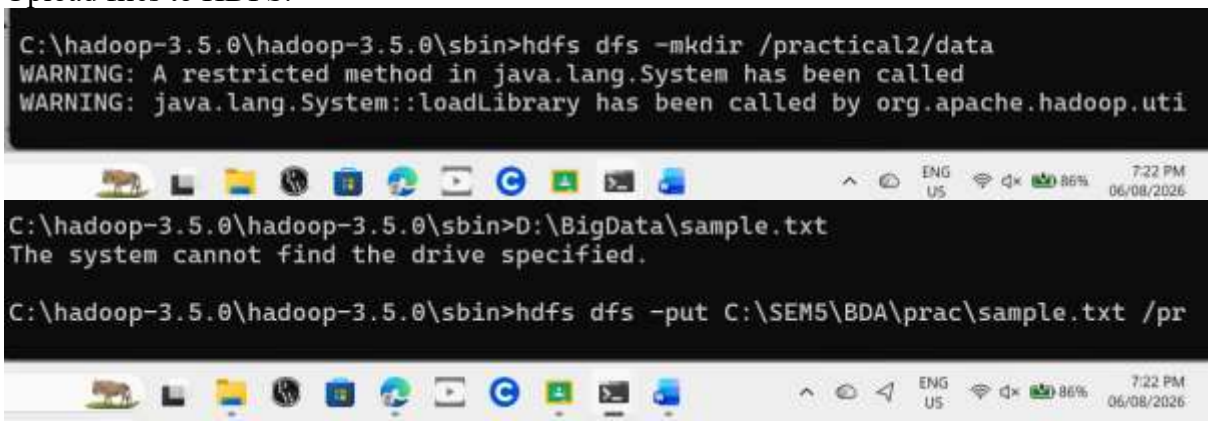
Access Hadoop Web UI.



1. Create directories in HDFS.



2. Upload files to HDFS.



```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -put C:\SEM5\BDA\prac\sample.txt /pr
actical2/data
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.uti
l.NativeCodeLoader in an unnamed module (file:/C:/hadoop-3.5.0/hadoop-3.5.0/sha
re/hadoop/common/hadoop-common-3.5.0.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers
```

3. List HDFS files and directories.

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -ls /practical2/data
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.uti
l.NativeCodeLoader in an unnamed module (file:/C:/hadoop-3.5.0/hadoop-3.5.0/sha
```

```
Found 1 items
-rw-r--r--   1 satya supergroup          0 2026-08-06 19:03 /practical2/data/sa
mple.txt

C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -cat /practical2/data/sample.txt
WARNING: A restricted method in java.lang.System has been called
```

4. Display file contents.

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -cat /practical2/data/sample.txt
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.uti
l.NativeCodeLoader in an unnamed module (file:/C:/hadoop-3.5.0/hadoop-3.5.0/sha
re/hadoop/common/hadoop-common-3.5.0.jar)
```

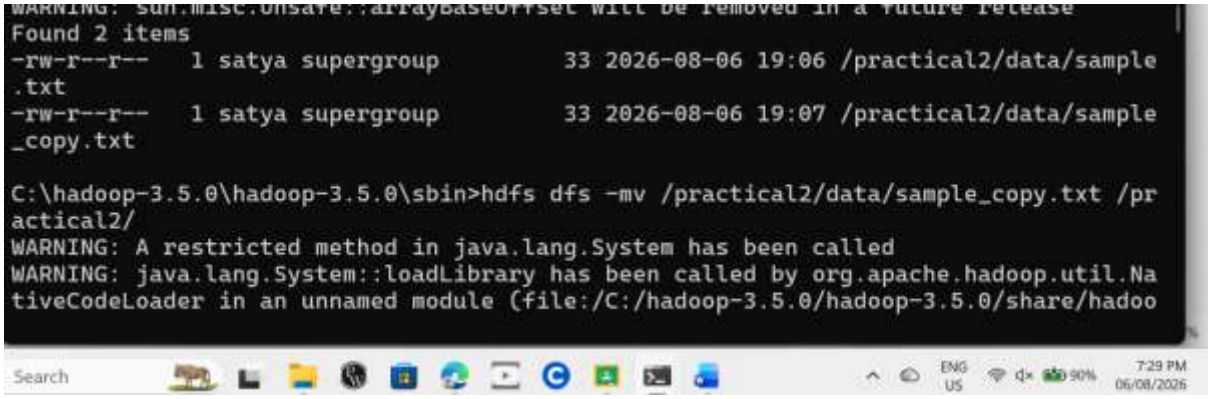
```
Hello, I am Satyarajsinh Rathod.
```

5. Copy files within HDFS.

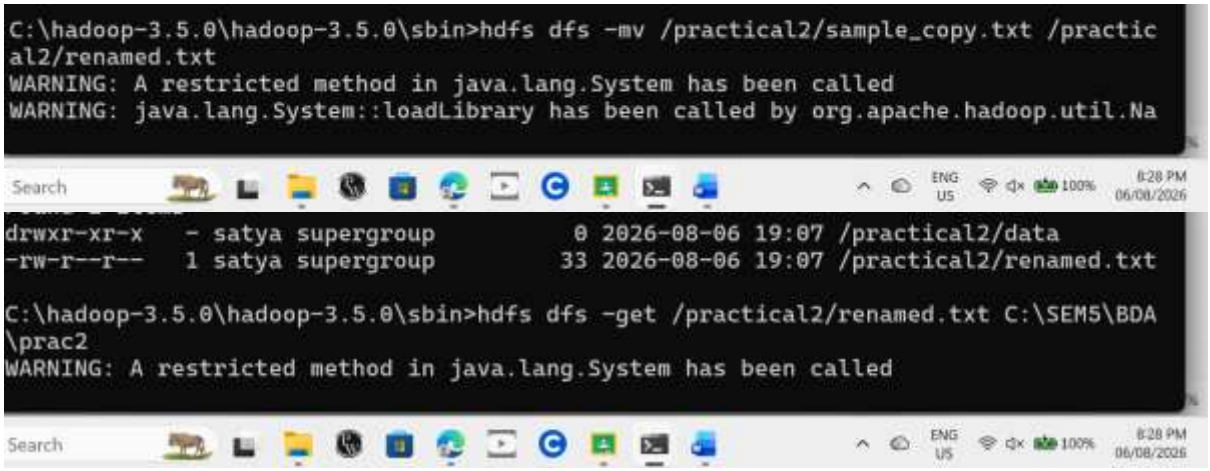
```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -cp /practical2/data/sample.txt /pra
ctical2/data/sample_copy.txt
WARNING: A restricted method in java.lang.System has been called
```

6. Move files within HDFS.

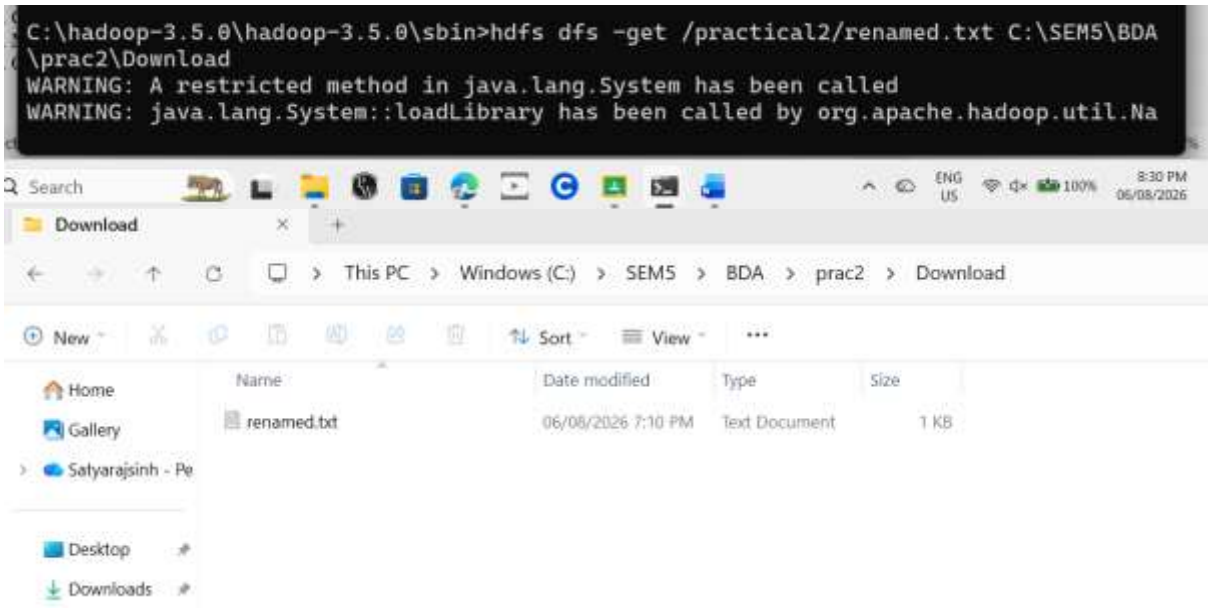
```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -cp /practical2/data/sample.txt /pra
ctical2/data/sample_copy.txt
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.uti
l.NativeCodeLoader in an unnamed module (file:/C:/hadoop-3.5.0/hadoop-3.5.0/sha
re/hadoop/common/hadoop-common-3.5.0.jar)
```



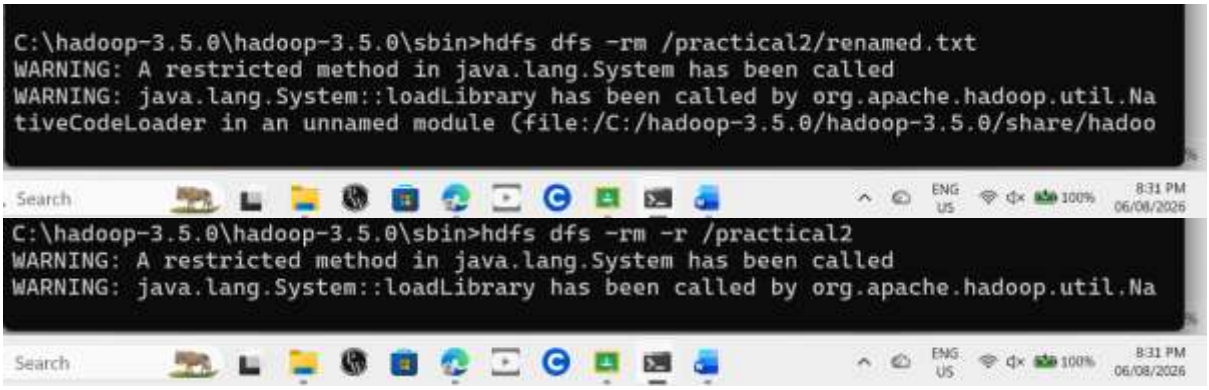
7. Rename files.



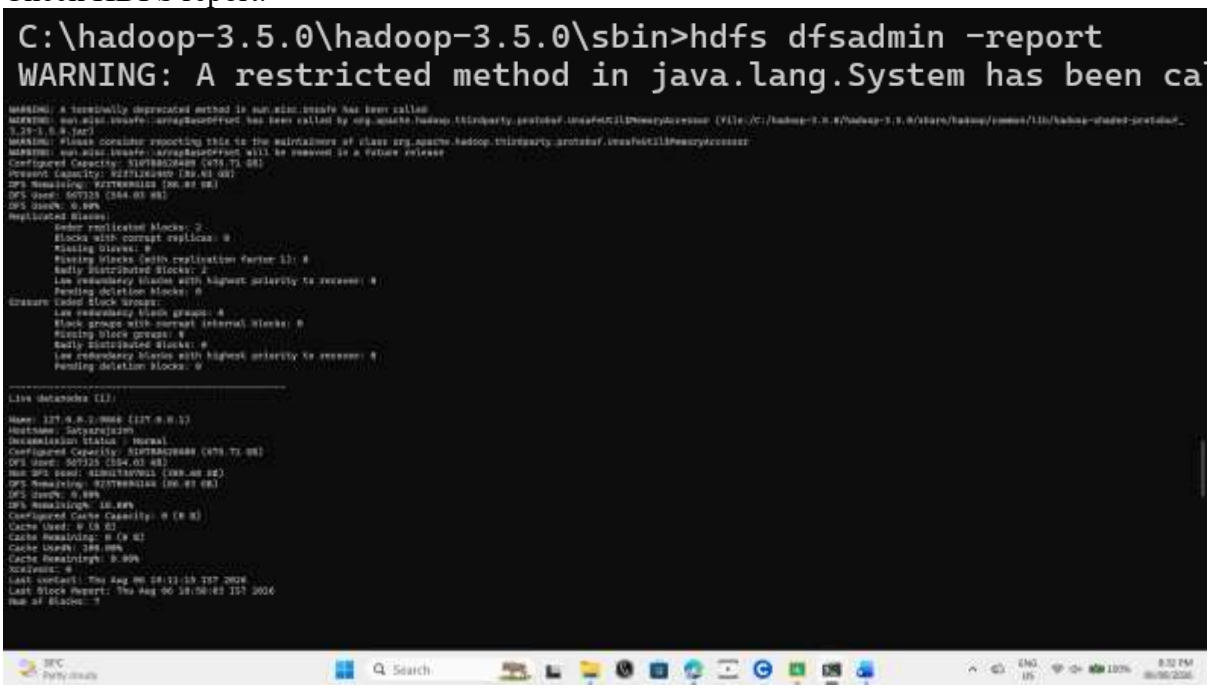
8. Download files from HDFS.



9. Delete files and directories.



10. Check HDFS report.



11. Verify block information.

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs fsck /
WARNING: A restricted method in java.lang.System
/tmp/hadoop-yarn/staging/satya/.staging/job_1784133662632_0001/job.sp
ecommissioned replica(s), 0 decommissioning replica(s).

Status: HEALTHY
 Number of data-nodes:  1
 Number of racks:               1
 Total dirs:                    9
 Total symlinks:                0

Replicated Blocks:
 Total size:    562534 B
 Total files:   7
 Total blocks (validated):      7 (avg. block size 80362 B)
 Minimally replicated blocks:   7 (100.0 %)
 Over-replicated blocks:        0 (0.0 %)
 Under-replicated blocks:       2 (28.571428 %)
 Mis-replicated blocks:         0 (0.0 %)
 Default replication factor:    1
 Average block replication:     1.0
 Missing blocks:                0
 Corrupt blocks:                0
 Missing replicas:              18 (72.0 %)
 Blocks queued for replication: 0

Erasure Coded Block Groups:
 Total size:    0 B
 Total files:   0
 Total block groups (validated):        0
 Minimally erasure-coded block groups:  0
 Over-erasure-coded block groups:       0
 Under-erasure-coded block groups:      0
 Unsatisfactory placement block groups: 0
 Average block group size:      0.0
 Missing block groups:          0
 Corrupt block groups:          0
 Missing internal blocks:       0
 Blocks queued for replication: 0
FSCK ended at Thu Aug 06 19:11:39 IST 2026 in 28 milliseconds
```

12. Check Hadoop cluster health.

```
The filesystem under path '/' is HEALTHY
```

13. Monitor Hadoop processes.

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>jps
24624 DataNode
26408 NameNode
20780 Jps
```

14. Check storage utilization.

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -du -h /
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.util.NativeCodeLoader in an unnamed module (file:/C:/hadoo
p-3.5.0/hadoop-3.5.0/share/hadoop/common/hadoop-common-3.5.0.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor (file:/C:
/hadoop-3.5.0/hadoop-3.5.0/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.5.0.jar)
WARNING: Please consider reporting this to the maintainers of class org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
15       15     /hello.txt
15       15     /input
15       15     /test
549.3 K  3.0 M  /tmp
```

Key Questions /Analysis:

1. **Why is Hadoop preferred for Big Data storage?**
   Hadoop is widely used for Big Data because it can store huge amounts of data on multiple computers. It is easy to expand by adding more machines, protects data by keeping copies, and provides an economical solution for handling large datasets.

2. **What is the role of NameNode and DataNode in HDFS?**
   The NameNode manages the HDFS system by keeping information about files, folders, and the location of data blocks. The DataNode stores the actual data blocks and sends them to users or applications whenever requested.

3. **What is the difference between HDFS and traditional file systems?**

HDFS	**Traditional File System**
Stores data across many computers	Stores data on one computer or disk
Suitable for very large datasets	Suitable for regular-sized files
Keeps multiple copies of data for safety	Usually stores only one copy unless backed up
Can grow by adding more machines	Expansion is limited by the machine's capacity
Built for distributed data processing	Built for everyday file storage

4. **What are the advantages of distributed storage?**
   Distributed storage allows data to be shared across multiple computers, making it easier to manage very large files. It increases reliability because data is copied to different machines, improves performance by processing data in parallel, and allows storage capacity to be increased without replacing existing hardware.

Conclusion:
The Hadoop single-node environment was successfully installed and configured on Windows. HDFS operations such as creating directories, uploading, managing, retrieving, and deleting files were performed successfully. The Hadoop cluster health, storage usage, and block information were also verified, demonstrating that HDFS can efficiently manage and store large-scale data.